Identifying Semantic Induction Heads to Understand In-Context Learning

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Abstract

Although large language models (LLMs) have demonstrated remarkable performance, the lack of transparency in their inference logic raises concerns about their trustworthiness. To gain a better understanding of LLMs, we conduct a detailed analysis of the operations of attention heads and aim to better understand the in-context learning of LLMs. Specifically, we investigate whether attention heads encode two types of relationships between tokens in natural languages: the syntactic dependency parsed from sentences and the relation within knowledge graphs. We find that certain attention heads exhibit a pattern where, when attending to head tokens, they recall tail tokens and increase the output logits of those tail tokens. More crucially, the formulation of such semantic induction heads has a close correlation with the emergence of the in-context learning ability of language models. The study of semantic attention heads advances our understanding of the intricate operations of attention heads in transformers, and further provides new insights into the in-context learning of LLMs.

1 Introduction

In recent years, the transformer-based large language models (LLMs) (Kaplan et al., 2020; Brown et al., 2020; Touvron et al., 2023; Bubeck et al., 2023) have rapidly emerged as one of the mainstreams in the field of natural language processing (NLP). While these models demonstrate emergent abilities as they scale (Brown et al., 2020; Wei et al., 2022), they become less interpretable due to the vast number of parameters and complex architectures, which emphasizes LLMs' safety and trustworthiness (Carlini et al., 2021; Manakul et al., 2023; Ren et al., 2024). Thus, beyond classical gradient-based explanations (Simonyan et al., 2013; Li et al., 2015), and perturbation-based explanations (Ribeiro et al., 2016; Lundberg and Lee, 2017; Sundararajan et al., 2017), recent studies in mechanistic interpretability (Cammarata et al., 2020; Elhage et al., 2021) attempt to reverse engineer the computations in transformers (particularly attention layers).

The mechanistic interpretability on transformer language models was first performed by Elhage et al. (2021). They disentangle two circuits from the operation of each attention head in transformers: Query-Key circuit (determines which token the head prefers to attend to) and Output-Value circuit (determines how the head affects the output logits of the next token). Then, Elhage et al. (2021) discover that some attention heads prefer to search for a previous occurrence of the current token in context and copy the next token associated with that occurrence, as shown in Figure 1. The attention heads performing such operations are termed *induction heads*. Taking a step further, Olsson et al. (2022); Bansal et al. (2023) have discovered that the presence of induction heads has a close correlation with the in-context learning (ICL) ability of LLMs. This finding highlights the importance of understanding the behavior of attention heads to the overall learning capabilities of LLMs.

On the other hand, semantic relationships have a vital importance on natural language understanding and processing. However, Elhage et al. (2021) only focus on whether the attention heads copy the attended token, without studying semantic relationships between tokens. Another major limitation of previous studies is that Olsson et al. (2022) does not explain the popular few-shot in-context learning schema. Instead, they study the loss decreasing along with the increase of token indices. This setting does not fully capture the complete ability of LLMs to learn from the context.

In this work, beyond simple copying, we delve deeper into high-level relationships encoded in at-

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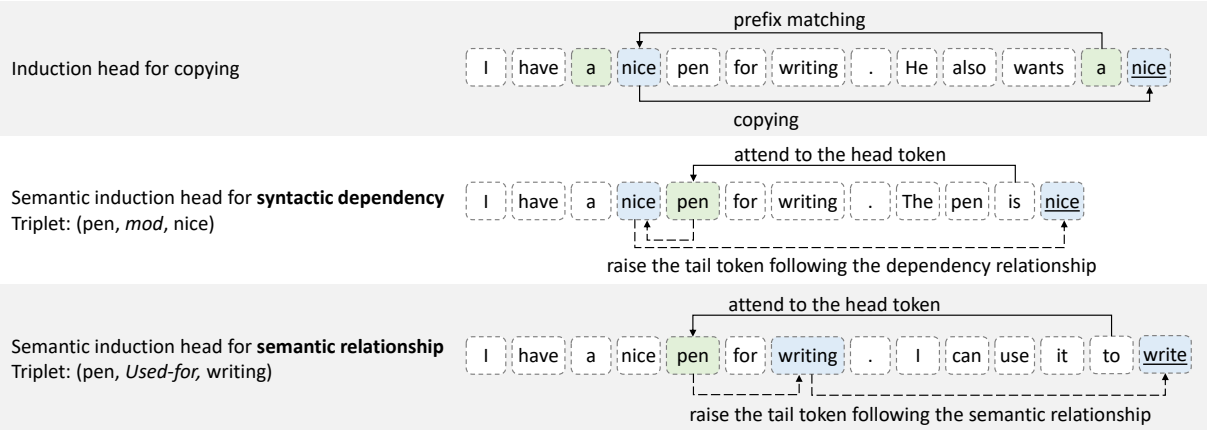

Figure 1: Induction heads and semantic induction heads. For the sequence "... a nice ... a", an induction head finds a place where the current token "a" occurred, attends to its next token "nice" (*prefix matching*), and then copies "nice" to the output *(copying*). In contrast, the semantic induction head raises the output logits of tail tokens ("nice" in the *mod* dependency and "writing" in the *Used-for* relation) when attending to the head token "pen".

tention heads. We focus on two types of relationships: (1) syntactic dependencies in the sentence and (2) semantic relationships between entities. Please refer to Figure 1 for examples. Each relation is represented as a triplet: (*head*, *relation*, *tail*). We find that when attending to head tokens, some attention heads prefer to raise the output logits of tail tokens associated with specific relations. Such attention heads encoding semantic relationships are termed *semantic induction heads*. Unlike conventional induction heads, semantic induction heads learn and leverage the semantic relationships between words to infer the output, thereby providing a better understanding of the behavior of networks.

Inspired by the study of induction heads and in-context learning, we further explore the correlation between semantic induction heads and in-context learning. We first categorize the in-context learning ability into three basic levels: loss reduction, format compliance, and pattern discovery. These three levels progressively increase in difficulty, with each subsequent level building upon the achievements of the previous one. The experimental results are consistent with our hypothesis, demonstrating the emergence of three levels of ICL in a sequential manner. Specifically, we observe the emergence of loss reduction from the beginning of the training, followed by the emergence of format compliance at around 1.6B tokens, and finally, the emergence of pattern discovery after training on approximately 4B tokens. Moreover, we find semantic induction heads mainly emerge around the same time as pattern discovery. Based on this finding, we infer that the emergence of semantic induction heads plays a crucial role in facilitating the ICL of LLMs.

Our contributions can be summarized as follows.

- We unveil the existence of semantic induction heads in LLMs that extract semantic relationships within the context. This discovery deepens the study of mechanistic interpretability and enhances our understanding of transformer-based models.
- To study the ICL in LLMs, we categorize it into three different levels and observe the gradual emergence of different levels of ICL during the early training stage of LLMs.
- Through a meticulous analysis of early checkpoints in the training of LLMs, we establish a close correlation between semantic induction heads and the occurrence of ICL.

2 Related Works

In this section, we provide an overview of recent advancements in the interpretability of neural networks, particularly mechanistic interpretability. On the other hand, previous studies (Petroni et al., 2019; Zhang et al., 2022) have also explored the topic of semantic relationships in models. However, our study distinguishes itself by focusing on mechanistic interpretability.

Previous studies in interpretability can be roughly categorized into the following four types: estimating the attribution of input features to the network output (Ribeiro et al., 2016; Sundararajan et al., 2017; Lundberg and Lee, 2017; Yang et al., 2023; Modarressi et al., 2023), discovering interaction patterns between input features (Ren et al., 2021, 2023; Liu et al., 2024; Zhou et al., 2024), extracting concepts from intermediate-layer

features (Kim et al., 2018; Thomas et al., 2023; Qian et al., 2024), and designing self-explainable architectures (Li et al., 2018; Das et al., 2022). As transformer-based models become mainstream, recent works focus on understanding the attention mechanism. The most direct approach is to visualize the attention using bipartite graphs (Vig, 2019; Yeh et al., 2024) or heatmaps (Park et al., 2019). Another line of research aims to reverse engineer the operation of attention heads, called mechanistic interpretability (Cammarata et al., 2020; Elhage et al., 2021).

Elhage et al. (2021) proposed the circuit analysis (introduced in Section 3) to examine the operation of attention heads, and they found induction heads in attention-only models. Olsson et al. (2022) further investigated the correlation between the formation of induction heads and ICL. Bansal et al. (2023) observed an overlap between the set of induction heads and the set of important attention heads for ICL. Using circuit analysis, Wang et al. (2023) also found some attention heads performing the function of identifying/removing names in the indirect object identification task. Other studies (Lieberum et al., 2023; Geva et al., 2023; Mohebbi et al., 2023) intervened the attention or FFN layers to study their functions. In this paper, we leverage the circuit analysis to investigate semantic relationships in attention heads.

3 Semantic Induction Head

Preliminary. Elhage et al. (2021) rewrite the operation of a multi-head attention (MHA) layer containing h attention heads as follows.

$$\begin{aligned}&\sum\nolimits_{h=1}^{H} \mathit{softmax}\left(\boldsymbol{x}W_q^h(\boldsymbol{x}W_k^h)^T/\sqrt{d_h}\right)\boldsymbol{x}W_v^hW_o^h\\ =&\sum\nolimits_{h=1}^{H} \mathit{softmax}\left(\boldsymbol{x}W_{QK}^h\boldsymbol{x}^T/\sqrt{d_h}\right)\boldsymbol{x}W_{OV}^h\end{aligned} \tag{1}$$

where $\boldsymbol{x} = [x_1^T, x_2^T, \ldots, x_N^T]^T \in \mathbb{R}^{N\times d}$ denotes the embedding sequence, and $x_i = t_iW_e \in \mathbb{R}^{1\times d}$ is the embedding of the i-th input token t_i. $W_e \in \mathbb{R}^{|\mathcal{V}|\times d}$ denotes the embedding layer over a vocabulary $\mathcal{V}$. $W_q^h, W_k^h, W_v^h \in \mathbb{R}^{d\times d_h}$ denote the query, key, and value transformations in the h-th attention head. The output transformation W_o can be decomposed as $W_o = [(W_o^1)^T\ (W_o^2)^T\ \ldots\ (W_o^H)^T]^T$, where $W_o^h \in \mathbb{R}^{d_h\times d}$.

In Equation (1), $W_{QK}^h = W_q^h(W_k^h)^T$, termed the Query-Key (QK) circuit, is responsible for computing the attention pattern of the head, thus determining the head prefers to attend to which token. On the other hand, the matrix $W_{OV}^h = W_v^hW_o^h$, termed the Output-Value (OV) circuit, computes the independent output of each head at the current token regardless of the attention pattern. The output of the OV circuit can be projected back to the vocabulary as $\boldsymbol{x}W_{OV}^hW_u$ by the unembedding transformation $W_u \in \mathbb{R}^{d\times|\mathcal{V}|}$. The projected vector represents the influence of the attention head on the output. Importantly, according to (Elhage et al., 2021), both the QK circuit and OV circuit are directly performed on input embeddings, facilitating the understanding of operations in attention heads.

Based on the above decomposition, Elhage et al. (2021) identify a specific behavior in attention heads, which they refer to as *induction heads*. They observe this behavior in attention heads when presented with sequences like "[A] [B] $\cdots$ [A]". In these induction heads, the QK circuit causes the attention head to attend to the token [B], which appears next to the previous occurrence of the current token [A]. This behavior is termed prefix matching. Then, the OV circuit increases the output logit of the attended token [B], termed *copying*. This mechanism is shown in Figure 1.

Main experimental setup. We use the open-sourced InternLM2-1.8B[1], which contains 24 layers and each layer consists of 16 attention heads. We use the Abstract GENeration DAtaset (AGENDA)[2] (Koncel-Kedziorski et al., 2019) for testing because it contains well-annotated relations between entities. The test set of the AGENDA dataset has a total of 1,000 samples, each consisting of a knowledge graph and a corresponding paragraph that describes relations in the knowledge graph. For syntactic dependencies, we split paragraphs in the AGENDA dataset into individual sentences, and use spaCy (Honnibal et al., 2020) to extract dependencies between tokens.

3.1 Syntactic Dependency in Attention Heads

In this section, we explore whether attention heads encode more complex knowledge beyond copying. We first study a basic pairwise relation inherent in natural language, syntactic dependency, representing the grammatical structure of a sentence.

We focus on three frequent types of syntactic dependencies: subject-predicate (*subj*), predicate-object (*obj*), and modifier-noun/verb (*mod*). Each relation is represented as a triplet: $T =$

[1] https://huggingface.co/internlm

[2] https://github.com/rikdz/GraphWriter